

Krishna Koushik Unnam

AI/ML Engineer

(510) 516-3735 | krishnakoushikunnam1@gmail.com | Menlo Park, CA | [LinkedIn](#) | [GitHub](#) | [Portfolio](#)

Summary

AI/ML Engineer with 5 years of experience designing, training, and deploying large-scale AI systems, including foundation models, multimodal LLMs, and enterprise machine learning solutions. Experienced in distributed model training, Generative AI, RAG, RLHF, MLOps, and scalable inference across GPU clusters and cloud platforms, delivering production-ready AI solutions for millions of users. Passionate about building intelligent systems that combine cutting-edge research with robust engineering to solve complex real-world problems at scale.

Professional Experience

Meta	Feb 2025 – Present
AI/ML Engineer	Menlo Park, CA
- Developed large-scale multimodal foundation model training pipelines using Python, PyTorch, Transformers, and distributed GPU infrastructure, processing trillions of tokens and improving benchmark accuracy by 18% across reasoning and coding evaluations. - Built Mixture-of-Experts (MoE) architectures using PyTorch, CUDA, FSDP, and Megatron-LM, optimizing expert routing and distributed training efficiency, reducing overall training time by 32% without compromising model quality. - Implemented RLHF, DPO, and preference optimization workflows using Python and PyTorch, enhancing response relevance, safety, and alignment while increasing human evaluation scores by 21% across production datasets. - Designed scalable data engineering pipelines using Spark, Ray, Airflow, and Python, enabling automated ingestion, cleansing, deduplication, and preparation of petabyte-scale multimodal and synthetic training datasets. - Developed retrieval-augmented generation solutions leveraging FAISS, embedding models, vector search, and transformer architectures, improving response accuracy by 24% for enterprise knowledge retrieval and contextual reasoning workloads. - Engineered comprehensive model evaluation frameworks integrating MMLU, GPQA, HumanEval, custom benchmarks, and automated reporting, accelerating validation cycles and ensuring deployment readiness for large foundation models. - Architected containerized AI microservices using Docker, Kubernetes, REST APIs, and service mesh technologies, supporting highly available inference platforms serving more than 600 million global users daily. - Deployed large-scale inference and model serving platforms across AWS, Azure, and Hyperscale private data centers, utilizing vLLM, TensorRT-LLM, and autoscaling strategies to reduce inference costs by 35%. - Implemented enterprise MLOps and CI/CD frameworks using MLflow, GitHub Actions, ArgoCD, Prometheus, and Grafana across GCP and Hyperscale private data centers, enabling continuous monitoring and governance. - Built agentic AI platforms on Azure, Kubernetes, and Hyperscale private data centers, integrating RAG, workflow orchestration, tool-calling, and distributed services to support scalable enterprise-grade intelligent applications.	
Accenture	Mar 2021 – Jun 2024
Machine Learning Engineer	India
- Developed Python-based predictive maintenance pipelines processing real-time IoT sensor data from over 15,000 industrial assets, enabling proactive failure detection strategies and reducing unplanned equipment downtime by 32%. - Built machine learning models using Python, TensorFlow, XGBoost, and historical maintenance datasets to predict equipment failures, improving prediction accuracy by 21% while supporting operational decisions for 5,000+ users. - Optimized real-time inference workloads through advanced feature engineering, model tuning, and MLflow experimentation, reducing prediction latency by 38% and lowering cloud inference costs by 27% annually. - Designed scalable data engineering workflows using Python, Apache Spark, Databricks, and SQL to ingest, process, and transform billions of manufacturing sensor records for downstream analytics. - Developed computer vision inspection solutions using OpenCV, PyTorch, and YOLO models to automate defect detection across production lines, significantly improving quality assurance efficiency and consistency. - Implemented end-to-end MLOps pipelines using MLflow, Git, Docker, Kubernetes, and CI/CD frameworks, enabling automated model training, deployment, monitoring, versioning, and governance across environments. - Architected cloud-native AI platforms on AWS utilizing Amazon S3, AWS Glue, Lambda, and SageMaker services, supporting secure data ingestion, feature engineering, model training, and deployment. - Built microservices-based prediction APIs using Java, Spring Boot, FastAPI, Docker, Kubernetes, and AWS EKS, delivering highly available real-time scoring services integrated with manufacturing execution and monitoring systems. - Developed digital twin analytics solutions leveraging AWS IoT Core, Databricks, and streaming telemetry data to simulate equipment behavior, predict failures, and optimize operational performance. - Created executive dashboards using Power BI, SQL, and predictive analytics outputs, providing actionable machine health insights, anomaly detection metrics, and maintenance forecasts to stakeholders.	

Technical Skills

Programming Languages: Python, SQL, Bash, Java, Go, C++, CUDA

AI/ML & Generative AI: PyTorch, TensorFlow, Scikit-learn, Hugging Face Transformers, LLMs, Multimodal AI, Foundation Models, Mixture of Experts (MoE), RLHF, DPO, SFT, AI Agents, Agentic AI, LangChain, LangGraph, LlamaIndex, MCP, Multi-Agent Systems, AI Agent Orchestration, Tool Calling, Function Calling, Workflow Automation, RAG, Embedding Models, Vector Search, Computer Vision, OpenCV, YOLO, XGBoost, LoRA, PEFT, Prompt Engineering, Knowledge Distillation

Distributed Training & Model Serving: PyTorch Distributed, FSDP, DeepSpeed, Megatron-LM, Ray, NCCL, vLLM, TensorRT-LLM, Model Quantization, GPU Optimization, Distributed Inference, HPC, Tensor Parallelism, Pipeline Parallelism, Data Parallelism

Data Engineering & Big Data: Apache Spark, Databricks, Apache Airflow, Apache Kafka, ETL Pipelines, Data Processing, Feature Engineering, Data Validation, Data Quality, Synthetic Data Generation, Streaming Data Pipelines, IoT Data Processing

Cloud Platforms & Infrastructure: AWS (S3, SageMaker, Glue, Lambda, EKS, IoT Core), Microsoft Azure, Google Cloud Platform (GCP), Kubernetes, Docker, Service Mesh, Infrastructure as Code (IaC), Terraform

MLOps, DevOps, Testing & Monitoring: MLflow, Git, GitHub Actions, CI/CD, ArgoCD, Pytest, Model Deployment, Model Versioning, Model Monitoring, Experiment Tracking, Prometheus, Grafana, OpenTelemetry, Kubernetes Operations, Feature Stores, AI Application Deployment

Backend, APIs & Application Development: FastAPI, Spring Boot, React, REST APIs, gRPC, Microservices, SSE

Evaluation, Databases, Vector Stores & Analytics: MMLU, GPQA, HumanEval, Custom Model Benchmarking, A/B Testing, PostgreSQL, MySQL, pgvector, Pinecone, Redis, FAISS, Power BI, Performance Optimization, Explainable AI (XAI), LangSmith

Professional Skills: System Design, Distributed Systems, Cross-Functional Collaboration, Technical Leadership, Analytical Thinking, Agile Development, Mentoring, Project Ownership, Documentation, Debugging, Innovation, Time Management, Continuous Learning

Projects

HELIX — PRODUCTION LLM INFERENCE GATEWAY | [GitHub](#)

Go, PostgreSQL, pgvector, Redis, Docker, Prometheus, Grafana, React

- Architected and deployed a multi-provider LLM inference gateway with score-based provider routing, pgvector semantic caching, JWT authentication with tenant-scoped controls, Redis token-bucket rate limiting, circuit-breaker fallback, SSE streaming, structured request logging, and Prometheus/Grafana observability.
- Load-tested the live, cache-warmed gateway across 3,885 requests and up to 100 concurrent virtual users, achieving 0% request failures, 18.43 requests per second, 567 ms p95 latency, and a 99.97% semantic-cache hit rate.

MULTI-AGENT CODE REPAIR PLATFORM | [GitHub](#)

Python, LangGraph, LiteLLM, GitHub API, tree-sitter, Docker, Pytest

- Built a stateful issue-to-pull-request workflow with repository research, minimal-fix planning, commit-pinned patch generation, Docker-isolated validation, structured retry feedback, and human escalation for ambiguous or unsuccessful repairs.
- Validated the workflow on two real open-source Click issues, passing 1,136–1,436 tests with clean Ruff and Mypy checks, zero retries, approximately two-minute execution time, and \$0.35–\$0.36 model cost per run.

DOCUMIND — EXPLAINABLE ENTERPRISE RAG PLATFORM | [GitHub](#)

Python, FastAPI, React, Pinecone, LangChain, LlamaIndex, OpenAI API

- Built an end-to-end document intelligence platform that extracts and chunks PDF content, generates embeddings, stores document scoped vectors in Pinecone, and serves context-grounded answers through an asynchronous FastAPI backend and React interface.
- Implemented document- and chunk-level metadata filtering, retrieved-source evidence, explicit empty-retrieval handling, and LlamaIndex relevancy evaluation to reduce cross-document contamination and make generated responses traceable to supporting context.

Education

Master of Science in Computer Science & Engineering

University of South Florida

Certifications

- AWS Certified Generative AI Developer – Professional (AIP-C01)
- Databricks Certified Machine Learning Professional
- Certified Kubernetes Application Developer (CKAD)
- Microsoft Certified: Azure AI Apps and Agents Developer Associate (AI-103)